

Supplemental Material for “Magnon–crystal-field cross peaks in two-dimensional terahertz spectroscopy of TmFeO₃”

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S1 The SU(3) qutrit and why its classical dynamics is quantum-exact

Tm³⁺ is $4f^{12}$, 3H_6 ($J = 6$). In the $C_s(m)$ crystal field of the $Pbnm$ $4c$ site the 13-fold multiplet splits into 13 singlets; the model keeps the three lowest, $|1\rangle, |2\rangle, |3\rangle$, with splittings $E_{12} = 2.068$ meV (0.50 THz) and $E_{13} = 4.963$ meV (1.20 THz), consistent with the measured lines $E_{12} + E_{23} \approx E_{13}$. The site state is the Gell-Mann vector $\lambda^a = \text{Tr}(\rho \Lambda^a)$, $a = 1 \dots 8$, and every single-ion term (CEF energies, external drives, exchange mean fields) is *linear* in the Λ^a . For a Hamiltonian linear in the generators the von Neumann equation $\dot{\rho} = -i[H, \rho]$ closes on $\vec{\lambda}$:

$$\dot{\lambda}^a = \Omega_{ab}(t) \lambda^b - \gamma_a (\lambda^a - \lambda_{\text{eq}}^a), \quad \Omega_{ab} = \frac{1}{2} \text{Tr} \left(\Lambda^a (-i) [H(t), \Lambda^b] \right). \quad (1)$$

The classical SU(3) propagation is therefore the *exact* quantum dynamics of the driven, damped three-level ion, and a Boltzmann-weighted ensemble of trajectories launched from the level seeds $|1\rangle, |2\rangle, |3\rangle$ is the exact thermal density matrix for single-ion drives. (This was verified explicitly against an independent density-matrix integrator; the two censuses agree to all digits quoted.) Damping is Bloch-type on the Gell-Mann pairs of each transition, $\gamma(\lambda^{1,2}), \gamma(\lambda^{4,5}), \gamma(\lambda^{6,7})$, with population relaxation $\gamma(\lambda^{3,8})$ toward the equilibrium populations captured at initialization.

The magnetically bright coordinate of the E_{12} block follows from the projected moment $PJ_zP = 5.264\lambda^2$: only λ^2 radiates magnetically, and the onsite splitting drives the planar precession $\dot{\lambda}^1 = -\Delta_{12}\lambda^2$, $\dot{\lambda}^2 = +\Delta_{12}\lambda^1$, so the E_{12} sector is a single driven oscillator

$$\lambda^2(t) = \int_0^t \sin[\omega_{12}(t-t')] h^1(t') dt', \quad (2)$$

where h^1 is the effective force a conversion vertex exerts on λ^1 . All delay-axis structure of a cross peak lives in the τ -dependence of h^1 .

S2 Crystallography and the projected Fe–Tm coupling

TmFeO₃ is $Pbnm$ ($Z = 4$): Fe³⁺ at $4b$ (site symmetry $\bar{1}$; every Fe sits on an inversion center), Tm³⁺ at $4c$ (site symmetry the single mirror m_z). Inversion maps each Fe sublattice to itself and exchanges Tm sublattices in pairs; the local mirror is the physical $4c$ site mirror and generates the relative λ^5, λ^7 signs between Tm sublattices. The Fe order is G-type ($T_N \approx 632$ K) with DM canting; below the spin reorientation ($T_2 \approx 85$ K) the ground state is $\Gamma_2 = (F_x, C_y, G_z)$.

The Fe–Tm sector is built from the physical operator route

$$\mathbf{S}, \mathbf{J} \longrightarrow H_{\text{Fe-Tm}} = \mathbf{S}^T \mathbf{A} \mathbf{J} \longrightarrow P \mathbf{J} P \longrightarrow \text{couplings to } \lambda^a, \quad (3)$$

never by assuming that a space-group rotation acts as a closed onsite SU(3) rotation in one fixed qutrit basis. Working in a *transported gauge* (qutrit bases on the four Tm sublattices related by the group operations themselves) makes every bond formula explicit and turns the irrep (Γ_1) projection into simple four-site sign sums. Time-reversal splits the Gell-Mann set into $\lambda^- = \{\lambda^2, \lambda^5, \lambda^7\}$ (odd) and $\lambda^+ = \{\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8\}$ (even); every coupling term must be $\mathcal{T}$ -even overall, which fixes how many Fe-spin or B -field legs may attach to each sector. The candidate couplings through cubic order in the fields are

family	operator form	sector	legs
K^-	$S_\alpha \lambda^a$	λ^-	1 Fe
ESJ	$E_\eta S_\alpha \lambda^a$	λ^-	1 E-photon + 1 Fe
κ^B	$B_\eta S_\alpha \lambda^a$	λ^+	1 B-photon + 1 Fe
W	$S_\alpha S_\beta \lambda^a$	λ^+	2 Fe

The full orbit formulas (all eight coefficients per bond, all four Tm sublattices, both time-reversal sectors) and the implementation prescriptions for global-axis versus local-frame storage are given in the long-form working note that accompanies the data (`tmfeo3.foundation.tex`, Secs. 3–14).

S3 The sharp requirement and the vertex scorecard

Two collinear pulses give $M_{\text{NL}} = M_{AB} - M_A - M_B$; order by order only the mixed terms survive, so M_{NL} starts at second order and the leading robust magnetic-dipole contribution is third order.

Combining Eq. (2) with the conservation rules at each vertex (energy; $\mathcal{T}$ -parity; Γ_1) yields six requirements for a coupling to produce the geometry-A cross peak at (q_{AFM}, E_{12}) :

R1 (detection) reaches λ^1/λ^2 (the E_{12} block);

R2 (drive) is sourced by the pumped δS_y magnon;

R3 (parity) is $\mathcal{T}$ -even (fixes the $\lambda^\pm$ sector);

R4 (irrep) is Γ_1 -allowed in the Γ_2 state;

R5 (energy) carries the $0.90 \rightarrow 0.50$ THz mismatch (needs ≥ 2 non- λ legs);

R6 (M_{NL}) involves both pulses.

Table S1: Vertex scorecard. Only W and κ^B pass all six requirements.

vertex	legs	sector	R1	R2–R6	verdict
K^-	$1\text{Fe} + 1\text{Tm}^-$	λ^-	$\times$	$\times$	hybridizes only; Γ_1 -blocked; reorients Γ_2
ESJ	$E + \text{Fe} + \text{Tm}^-$	λ^-	$\times$	$\checkmark$	real, but feeds E_{13}/E_{23} , not E_{12}
W	$2\text{Fe} + \text{Tm}^+$	λ^+	$\checkmark$	$\checkmark$	winner: rank-1, factorized peak
κ^B	$B + \text{Fe} + \text{Tm}^+$	λ^+	$\checkmark$	$\checkmark$	works; B_z -gated $\Rightarrow$ geometry-B vertex

K^- fails threefold. (i) Linear in S , it only rotates the harmonic normal-mode basis: coupled harmonic oscillators give diagonal peaks only; a static bilinear conserves energy and cannot convert 0.90 into 0.50 THz. (ii) The Γ_1 orbit sums vanish for every channel except K_{2y}^- : a static one-at-a-time scan at 0.2 meV leaves the energy, the Fe order, and the uniform $(\lambda^2, \lambda^5, \lambda^7)$ *bit-identical* to baseline for all components except $2y$. (iii) The one allowed channel K_{2y}^- is a transverse molecular field on the order parameter and reorients $\Gamma_2 \rightarrow G_y$ between 0.085 and 0.090 meV—it is not a perturbative vertex. Full-dynamics 2DCS on the inert channels (K_{2x}^- up to 1.5 meV, K_{2z}^- up to 2.0 meV) leaves the induced λ^2 at machine precision ($\leq 7 \times 10^{-15}$).

The two winners. The two-magnon vertex rectifies the pulse-A/pulse-B cross term,

$$h_{\text{NL}}^1(t, \tau) = W_1^{yy} A^2 [\cos(q_{\text{AFM}}\tau) - \cos(q_{\text{AFM}}(2t + \tau))] \theta(t) \Rightarrow M_{\text{NL}}^W \propto \cos(q_{\text{AFM}}\tau) [1 - \cos \omega_{12}t], \quad (4)$$

a clean factorized peak with no diagonal partner. The field-gated vertex kicks at the probe, $h^1 \propto B_y(t)S_y(t)$ with $S_y(0) = A \sin q_{\text{AFM}}\tau$, giving $M_{\text{NL}}^K \propto \sin(q_{\text{AFM}}\tau)[1 - \cos \omega_{12}t]$: also on target, but the same gate dumps each pulse’s local response into Tm at both pulse times, generating an (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background. Its dominant symmetry slice is z -gated ($B_z(t)$), so in geometry A ($B_z = 0$) it is *identically silent*: the gate is simultaneously the reason it is harmless in geometry A and active in geometry B.

Exhaustion of the quadratic families. All remaining symmetry-allowed quadratic couplings were switched on one at a time in the full nonlinear dynamics and are exactly inert (differences at the 10^{-12} level or below): $W_4^{yy,xy,yz,xx,zz,xz}$, W_1^{xy} , $W_6^{yy,xz}$, κ_{2y}^E , κ_{5x}^E . Among the W_1 components: W_1^{yz} inert, W_1^{zz} negligible, W_1^{xx} alive but floods the spectrum with an $\omega_t = 0.16$ THz comb, W_1^{xz} alive and on-target for hybridization (used below), and no static component can replace the gated κ^B for the geometry-B transfer peak: the entire W_1+W_4 family fails to produce (E_{12}, q_{AFM}) in geometry B without simultaneously contaminating geometry A. The B_z -gate is therefore *proven necessary by exhaustion*.

S4 Numerical protocol

Dynamics. Classical LLG for Fe (unit spins, $S = 5/2$ scale absorbed), SU(3) Bloch dynamics for the four Tm sublattices; RK4 with $\Delta t = 0.02$ code units; $1 \times 1 \times 1$ cell of the *Pbnm* magnetic unit (8 Fe + 4 Tm). Time is measured in $\hbar/\text{meV}$; a frequency f in THz corresponds to $E = f \times 4.136 \text{ meV}$.

2DCS protocol. Delay scan $\tau \in [-120, 0]$ code units, $\Delta\tau = 0.1$ (0.2 for scans); detection window to $t_{\text{max}} = 160$ for 0.01 THz resolution on ω_t . The nonlinear signal is the honest three-run subtraction $M_{\text{NL}} = M_{AB} - M_A - M_B$ per delay; no pulse-window chunking; no reuse of the reference run across delays (both shortcuts were found to contaminate spectra and are documented pitfalls). The pump enters as a tabulated waveform (the measured field, times converted $t_{\text{code}} = t_{\text{ps}}/0.6582$, auto-centered and normalized).

Spectra and blind peak census. 2D spectra are $|\text{FFT}_{\tau,t}|$ of M_{NL} with: detection window $t \geq 3$ (excludes pulse overlap), Gaussian t -apodization to 0.03, cosine tapers on both τ edges, global mean subtraction. The delay axis is plotted physically ($\omega_T = -\omega_T^{\text{FFT}}$; $+$ = non-rephasing). Peaks are accepted only by a blind census: 2D local maxima (maximum filter, 0.10 THz neighborhood) with prominence ≥ 1.5 over the local median background (0.34 THz annulus) and amplitude ≥ 4 –5% of the map maximum, merged within 0.09 THz. Rectified ($\omega_t \approx 0$) content is analyzed separately via $S_{\text{DC}}(\tau) = \langle M_{\text{NL}} \rangle_t$ with cubic drift removal.

Thermal ensembles. Boltzmann mixtures of trajectories from the level seeds $|1\rangle, |2\rangle, |3\rangle$ (Sec. S1). Two documented pitfalls: (i) any pre-dynamics energy minimization collapses excited-level seeds and must be disabled; (ii) population damping $\gamma(\lambda^{3,8})$ relaxes toward the equilibrium captured at initialization, which must therefore be captured per-seed.

S5 Experimental inputs

Table S2: Measured mode parameters and the derived simulation inputs. CEF damping is Bloch on the Gell-Mann pair of each transition, $\gamma_a[\text{meV}] = \gamma[\text{THz}] \times \hbar$; magnon damping is Gilbert, $\alpha = \gamma/2\nu$ fixed by the q_{AFM} line. Emission weight is $\sqrt{f}$ (E1) for the CEF lines and M1 for the magnons. The last column is the measured pulse spectral amplitude at each mode.

mode	ν [THz]	f	$\sqrt{f}$	γ [THz]	damping (sim)	pulse amp.
q_{FM}	0.38	0.08	0.28	0.005	$\alpha = 1.7 \times 10^{-3}$	0.92
q_{AFM}	0.90	7×10^{-4}	0.026	0.003	$\alpha = 1.7 \times 10^{-3}$	0.68
E_{12}	0.50	6.5	2.55	0.038	$\gamma(\lambda^{1,2}) = 0.157 \text{ meV}$	1.00
E_{23}	0.70	6.6	2.57	0.10	$\gamma(\lambda^{6,7}) = 0.414 \text{ meV}$	0.85
E_{13}	1.20	8.7	2.95	0.05	$\gamma(\lambda^{4,5}) = 0.207 \text{ meV}$	0.36

Two consequences organize the readout physics. First, excitation scales as E^2 and detection as E^1 (third order). Second, the CEF lines are E1-bright while q_{AFM} is E1-dark but M1-bright: a magnon is therefore read out either by borrowing a CEF electric dipole through an Fe–Tm vertex (geometry A) or through its own magnetic dipole at the magnon frequency (geometry B). Using the magnetic g -factor instead of $\sqrt{f}$ for the magnon electric emission over-weights it by $\sim 100\times$ and buries the CEF peaks under the magnon diagonal; the oscillator-strength weighting resolves this without any tuning.

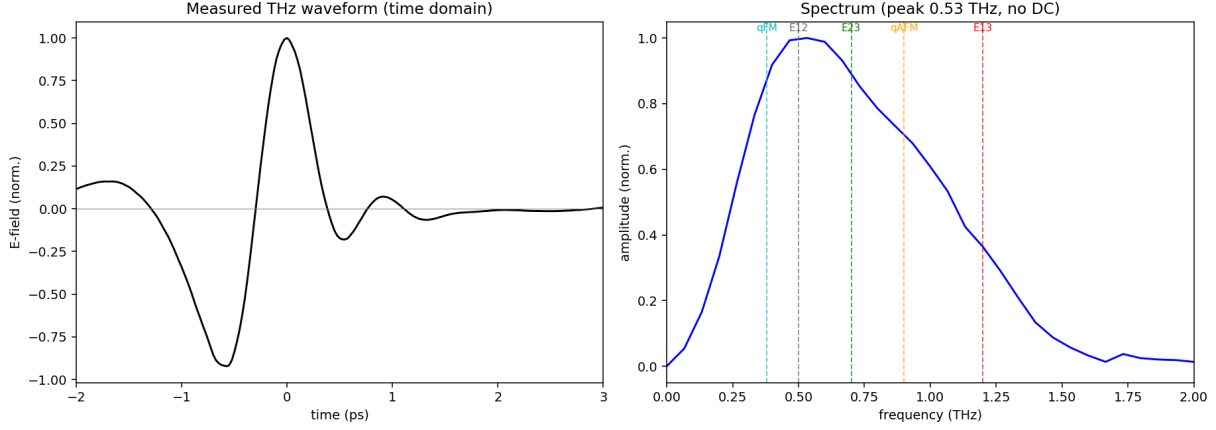


Figure S1: The measured THz pump. Left: single bipolar cycle (~ 1.2 ps, no DC). Right: amplitude spectrum peaked at $0.53 \text{ THz} \approx E_{12}$, covering all five modes. The absence of DC content removes the quasistatic-tilt legs that a Gaussian stand-in pulse would add; the measured waveform is the cleanest configuration.

S6 Displacive verification of the W mechanism

The rectified two-magnon force is a suddenly switched step: its edge carries spectral weight $\propto e^{-\omega_{12}^2 \sigma^2 / 2}$ at the CEF frequency, i.e. the E_{12} quantum is drawn from the pulse bandwidth (intra-pulse difference-frequency/impulsive-Raman), while the magnon pair carries only the phase memory $\cos q_{\text{AFM}} \tau$. Verified directly: stretching the pulse $\sigma = 0.25 \rightarrow 1.0$ at fixed amplitude collapses the peak by 4.5 decades ($51.7 \rightarrow 1.7 \times 10^{-3}$), while switching the direct Tm drive on/off changes it by $< 10^{-4}$ (pure- W origin). The same sweep resolves the second Fe leg: impulsive pulses use the resonant probe magnon (peak $\propto A^2$); longer pulses cross over to the field-following quasistatic tilt (peak $\propto A^1$). The measured no-DC waveform suppresses the quasistatic legs, which is why the experimental pulse yields the cleanest spectra.

S7 Geometry B: gate versus buildup

Isolated-vertex runs: κ_{1y}^B alone $\Rightarrow (E_{12}, q_{\text{AFM}}) = 0.46$ present, no buildup; W_1^{xz} alone $\Rightarrow$ peak absent, buildup present. Both retained at 0.01 code units. In geometry B the Tm CEF channels λ^{4-7} are *exactly* zero (machine precision): the $E \parallel a, H \parallel c$ drive closes on the $\{\lambda^1, \lambda^2, \lambda^3\}$ subalgebra, an $\text{SU}(2)$ closure that protects geometry B from CEF contamination.

Finite temperature: $(q_{\text{FM}}, q_{\text{AFM}})$ is a pure two-magnon process and temperature-flat; $(E_{12}, q_{\text{AFM}}) \propto n_1 - n_2$ falls $0.43 (0 \text{ K}) \rightarrow 0.41 (5) \rightarrow 0.35 (8) \rightarrow 0.29 (10 \text{ K})$. Both peaks remain resolved through $\sim 8 \text{ K}$, defining the working window 5–8 K.

S8 The same-polarization channel: pathways, exact single-ion verification, and the falsification ladder

S8.1 Configuration

The final same-polarization configuration (run `it14hr`; parameter files in `tmfeo3_2dcs_final/params_samepol_final`) is the unchanged v4 Hamiltonian with: measured pulse; drive vector $(0, 3.0, 0, 0, \mathbf{0}, 0, 0.9128, 0)$ on $(\lambda^1, \dots, \lambda^8)$ —i.e. the $1 \rightarrow 3$ leg removed ($\mu_{13}^z = 0$ in drive); per-leg amplitude twice the weak-probe

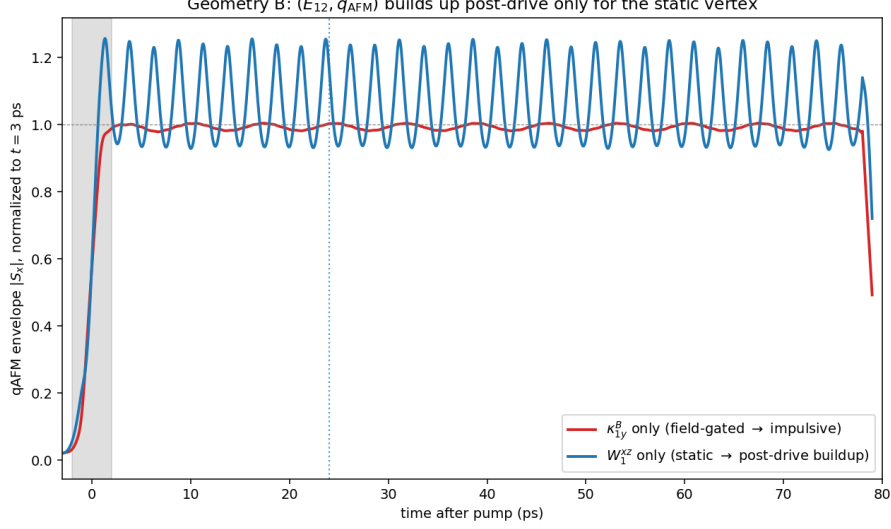


Figure S2: The q_{AFM} envelope in geometry B, normalized just after the pump. The field-gated κ_{1y}^B (which alone produces the (E_{12}, q_{AFM}) cross peak) leaves a flat post-pulse envelope (modulation 0.03); the static W_1^{xz} (which alone cannot produce that peak) keeps exchanging the q_{FM}/E_{12} reservoir with the magnon (modulation 0.33, envelope $\sim 25\%$ above the impulsive level). The measured post-drive buildup requires the static vertex; the measured transfer peak requires the gated one.

baseline; dampings of Table S2 plus population relaxation $\gamma(\lambda^{3,8}) = 0.08$; Boltzmann ensemble; detection composite $0.05\lambda^4 + 4.4\lambda^6$ ($\mu_{13}^z/\mu_{23}^z \approx 1\%$ in emission), magnon M1 excluded per Table S2.

S8.2 Pathway assignments

With $\mu_{13}^z = 0$ the Liouville pathways of the five listed peaks are:

peak (ω_T, ω_t)	model (10 K)	side	pathway
(+0.49, 0.72)	1.00	NR	$\rho_{12} \xrightarrow{\text{probe } 1 \leftrightarrow 2} n_2(\tau\text{-phase}) \xrightarrow{\text{probe } 2 \leftrightarrow 3} \rho_{23}$
(0.5, 0)	sharpest S_{DC} line	—	same storage, zero-frequency output leg
(+0.49, 1.20)	0.78	NR	$\rho_{12} \xrightarrow{\text{probe } 2 \leftrightarrow 3} \rho_{13}$, emission throttled by μ_{13}^z
(+0.91, 1.18)	0.20 $\rightarrow$ 0.51 (10 $\rightarrow$ 20 K)	NR	q_{AFM} delay, Fe–Tm handoff, thermally assisted
(0.9, 0.38)	absent	—	carrier exists (b -axis M1 q_{FM} ridge); no $q_{\text{AFM}} \rightarrow q_{\text{FM}}$ vertex
(−0.47, 0.70)	0.55	R	rephasing remnant (exact single-ion floor; prediction)
(+0.99, 0.71)	0.49	NR	inter-site $2E_{12}$ two-quantum line (prediction)

S8.3 Exact single-ion verification

A standalone integrator of the 8-dimensional $\vec{\lambda}$ equation (same CEF energies, dampings, measured pulse, same census pipeline; ~ 13 s per 2D map versus ~ 10 min for the lattice) reproduces the lattice 0.5-family census exactly, proving the family is single-ion physics and enabling exhaustive scans. Two facts emerge: (i) the rephasing/non-rephasing ratio of the (E_{12}, E_{23}) transfer peak has a floor of 0.47–0.58 under this pulse across all drive strengths, dipole-phase choices, and dampings—the rephasing remnant is an exact prediction of any three-level ion driven by this waveform; (ii) the

$\omega_\tau = 2E_{12}$ two-quantum line is absent in the exact single-ion dynamics and is therefore an inter-site (mean-field) product.

S8.4 Falsification ladder

The dark-dipole scenario was reached by exhausting the alternatives; each row was run in the full lattice model with the blind census:

hypothesis	run	outcome
static mixing field $h_{\lambda^6} = 0.3$	it8	over-strong: transition lines shift off calibration
perturbative $h_{\lambda^6} = 0.06$	it9	pumps (q_{AFM}, E_{13}) to 0.9 (wrong direction); side unchanged
CEF detuning 0.02 THz (ladder break)	it10	interloper survives; larger detuning barred by the observed degenerate lines
dipole triple-product sign flip	it11	census unchanged
dark μ_{13} at low drive	it12	population route floods the $(0, E_{23})$ band; needs the $2\times$ drive + T_1 combination
dark μ_{13} , $2\times$ drive, T_1	it14 adopted:	both transfer peaks NR, correct hierarchy
negative μ_{13} leg, lower drive	it15	interloper returns ($ \mu_{13} $ acts sign-independently); magnon-delay family floods

The $(0.9, 0.38)$ peak remains the one open item: the q_{FM} emission carrier exists as a uniformly τ -modulated b -axis M1 ridge in the Fe S_y channel (Γ_2 puts $\delta F_{y,z}$ at q_{FM}), but the model lacks a $q_{\text{AFM}} \rightarrow q_{\text{FM}}$ transfer vertex of sufficient strength; one probed configuration (it15) produced a genuine $(+0.89, 0.33)$ maximum at 0.21 relative, evidence the peak is reachable in an extended Hamiltonian class.

S9 Data and code availability

All parameter files, the blind-census scripts, the standalone qutrit integrator, and the long-form derivation note (`tmfeo3_foundation.tex`: transported-gauge orbit formulas, vertex diagrams, complete written-out Hamiltonian, mode inventory) are archived with the repository (`tfo_project/`); every spectrum in the paper can be regenerated from the committed `.param` files with the public `ClassicalSpin.Cpp` solver.